Fourier Transforms with Potential Fields
Practical Assignment

Geophysics IIIA

Overview

In this practical you will use Fourier transforms to analyse gravity and magnetic data from South Australia. You will explore how spectral methods relate wavelength to source depth, how continuation filters modify potential-field data, how isostatic compensation can provide a physically motivated regional field, and how reduction to the pole corrects for the inclination of the Earth's magnetic field.

Complete the accompanying Jupyter Notebook and answer the interpretation questions below. Your responses should demonstrate physical understanding rather than simply restating computational steps.

Learning Objectives

By completing this assignment, you should be able to:

- Interpret potential-field data in both spatial and spectral domains
- Explain upward and downward continuation in physical and mathematical terms
- Separate regional and residual fields and critically evaluate the choices involved
- Compare spectral and isostatic approaches to defining a regional gravity field
- Apply reduction to the pole and understand its assumptions and limitations
- Discuss how sediment thickness affects the interpretation of magnetic data
- Recognise the non-uniqueness inherent in potential-field interpretation

1 One-Dimensional Profile Continuation

Gravity and magnetic profiles extracted from the gridded data are continued upward and downward using a spectral filter. The continuation filter is:

$$H(k, z) = e^{-2\pi|k|z}$$

where k is wavenumber (cycles km^{-1}) and $z > 0$ is upward continuation height in km.

Question 1 — Upward continuation as a low-pass filter

As you increase the continuation height, which features disappear first — broad, regional anomalies or narrow, localised ones? What does this tell you about the relationship between anomaly wavelength and source depth?

Question 2 — Residual anomalies

What geological features might the residual profile (observed – regional) represent? Are the residuals in the gravity and magnetic profiles spatially correlated along your chosen profile? What could explain agreement or disagreement between the two fields?

Question 3 — Choosing a continuation height

There is no single "correct" continuation height for separating regional from residual.

a. What criteria might you use to make a defensible choice of continuation height?

b. How sensitive is your geological interpretation to that choice?

Question 4 — Instability of downward continuation

Move the continuation-height slider into negative values (downward continuation).

a. At approximately what depth does the downward-continued field become visibly unstable?

b. Does instability set in at the same depth for gravity and magnetics, or does one break down sooner?

c. Why might they differ?

Question 5 — Physical meaning of the continued field

Upward continuation produces the field that *would* be observed at a higher elevation. Does the continued profile look like what you would expect from a smoother, more distant source distribution? Are there any features that seem physically unreasonable, and if so, what might cause them?

2 Two-Dimensional Gravity Continuation and Isostatic Correction

The 2-D continuation filter extends the 1-D result to two spatial frequencies:

$$H(k_x, k_y, z) = e^{-2\pi\sqrt{k_x^2+k_y^2}\,z}$$

An isostatic correction based on Airy compensation provides an alternative, geologically motivated regional field. Under Airy isostasy, topographic load h is compensated by a crustal root of thickness $t_r = \rho_c\, h\,/\,(\rho_m - \rho_c)$, producing a gravity effect:

$$g_{\text{iso}}(x, y) = 2\pi G \rho_c\, h(x, y)$$

Question 6 — Spectral regional vs isostatic correction

Compare the spectral regional field (at your chosen continuation height) with the isostatic correction map.

a. How well do they agree spatially?

b. Which areas show the largest differences, and why?

Question 7 — Residual comparison

At what continuation height does the spectral residual most closely resemble the isostatic residual? What does this imply about the effective depth of isostatic compensation in South Australia?

Question 8 — Anomalies not explained by isostasy

What geological features might explain anomalies that are present in the spectral residual but absent (or reduced) in the isostatic residual?

Question 9 — Limitations of the isostatic model

What assumptions in the Airy isostasy model most strongly limit its accuracy in this region? Consider both the model geometry and the physical properties used.

3 Reduction to the Pole

Reduction to the pole (RTP) applies a phase correction that transforms total magnetic intensity (TMI) data as though the ambient field and induced magnetisation were both vertical. The filter is:

$$H_{\mathrm{RTP}}(k_x, k_y) = \frac{\bar{\Theta}^2}{\left(|\Theta|^2 + \varepsilon\right)^2}, \quad \Theta = \sin I + i \cos I \left(\frac{k_x \cos D + k_y \sin D}{|\mathbf{k}|} \right)$$

where I and D are the geomagnetic inclination and declination.

Question 10 — Effect of RTP on anomaly positions

How does reduction to the pole change the positions and shapes of the major magnetic anomalies in South Australia? Are anomalies shifted northward or southward, and by roughly how much?

Question 11 — Stability at low inclinations

Experiment with the inclination slider.

a. At what inclination do artefacts (ringing, extreme amplitudes) begin to appear?

b. What does this tell you about the reliability of RTP near the magnetic equator?

Question 12 — Difference map

Examine the difference map (RTP − TMI).

a. How does it relate to the asymmetry of individual anomalies in the TMI?

b. What geological information might be encoded in that difference?

Question 13 — Remanent magnetisation

The RTP filter assumes purely *induced* magnetisation parallel to the ambient field. How would remanent magnetisation with a different direction affect the result? Can you identify any anomalies in the dataset that might carry significant remanence?

4 Sediment-Thickness Correction for RTP [*Optional*]

Thick, non-magnetic sedimentary cover attenuates basement magnetic signals. A spectral correction attempts to undo this attenuation by multiplying the spectrum by $e^{+2\pi|\mathbf{k}|z_s}$, where z_s is the sediment thickness. A Gaussian taper $e^{-2\pi^2|\mathbf{k}|^2\lambda^2}$ damps the resulting noise amplification.

Question 14 — Where the correction matters most

Where does the sediment correction make the largest difference to the RTP result? Does this correspond to the areas of thickest sediment shown in the data?

Question 15 — Role of the damping parameter

What happens when you increase the damping length λ? What signal is being suppressed, and is this geologically sensible?

Question 16 — Validity of a uniform correction

The correction here uses the mean sediment thickness across the study area.

a. In which parts of the study area do you expect this uniform approximation to be most and least valid?

b. What additional information would you need to make the correction spatially variable in a rigorous way?

Question 17 — Analogous gravity correction

Could a similar sediment-correction approach be applied to the gravity field? How would it differ from the Airy isostatic correction computed in Section 2?

5 Synthesis

Question 18 — Wavelength and depth

How do Fourier-domain methods relate signal wavelength to source depth? State the approximate relationship, identify its key assumptions, and describe one situation in which it would give a misleading depth estimate.

Question 19 — Spectral vs isostatic regional field

What are the advantages and disadvantages of using spectral continuation versus isostatic correction to separate regional from residual gravity? Under what geological conditions would you prefer one over the other?

Question 20 — Non-uniqueness

All potential-field methods suffer from the fundamental non-uniqueness of the inverse problem. Give one concrete example from this practical where two different assumptions produced similar-looking results. What additional data or constraints could help discriminate between them?

Submission Instructions

- Provide written answers to the questions above, either within the notebook or as a separate document as specified by your instructor.
- Include screenshots/figures from your Jupyter Notebook as necessary to answer the questions.
- Figures should be clearly labelled and referred to in your written answers where appropriate.

Assessment Criteria

Your work will be assessed on the basis of:

- Physical understanding of Fourier methods and potential fields
- Correct and insightful interpretation of results
- Clarity and quality of written explanations
- Appropriate use of figures to support interpretation

End of Assignment